

SAT Math Mastery · Week 8B

Circles: Equations, Arcs, Tangents & Radians

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics circle equation · completing the square · center/radius from expanded form · circle position tests · arc length & sector area · central & inscribed angles · tangent $\perp$ radius · radians & degrees · unit circle

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Read the center and radius from $(x - h)^2 + (y - k)^2 = r^2$, and write the equation from a center and radius, a diameter, or a graph.
- S2.** Complete the square to convert $x^2 + y^2 + ax + by + c = 0$ to standard form; use $r^2 > 0$ to find a constant.
- S3.** Translate or enlarge a circle and track the center and r^2 .
- S4.** Decide whether a point is on, inside or outside a circle; use tangency to an axis ($|\text{center coordinate}| = r$) or to another circle (distance between centers $= r_1 + r_2$ or $|r_1 - r_2|$).
- S5.** Compute arc length and sector area from a central angle in degrees or radians, and work backward to the radius or angle.
- S6.** Use the central-angle and inscribed-angle theorems (inscribed angle $= \frac{1}{2}$ central angle on the same arc; angle in a semicircle is 90° ; opposite angles of an inscribed quadrilateral are supplementary).
- S7.** Use “a tangent is perpendicular to the radius at the point of tangency”, including equal tangent segments from an outside point.
- S8.** Convert between radians and degrees; evaluate $\sin$, $\cos$, $\tan$ at standard angles on the unit circle and use quadrant signs.

2 · REQUIRED FACTS

- The graph of $(x - h)^2 + (y - k)^2 = r^2$ is a circle with center (h, k) and radius r (the right side is r^2 , not r).
- A point is inside the circle if its distance to the center is less than r , on it if equal, outside if greater.
- A circle is tangent to the x -axis when $|k| = r$, and to the y -axis when $|h| = r$.
- A full circle is $360^\circ = 2\pi$ radians, so π rad $= 180^\circ$. An arc is the fraction $\frac{\theta}{360^\circ}$ (or $\frac{\theta}{2\pi}$) of the whole circle.
- An inscribed angle is half the central angle that intercepts the same arc. An angle inscribed in a semicircle is a right angle.
- A tangent line is perpendicular to the radius drawn to the point of tangency; two tangent segments from one outside point are equal.
- On the unit circle the point at angle θ is $(\cos \theta, \sin \theta)$; the signs follow the quadrant (All, Sine, Tangent, Cosine going counterclockwise).
- Common values: $\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{\sqrt{2}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$; $\cos$ is the same list in reverse order.

3 · REQUIRED FORMULAS

$$(x - h)^2 + (y - k)^2 = r^2, \quad x^2 + y^2 + ax + by = c \Rightarrow \left(x + \frac{a}{2}\right)^2 + \left(y + \frac{b}{2}\right)^2 = c + \frac{a^2 + b^2}{4}$$

$$s = r\theta \text{ (}\theta \text{ in radians)} = \frac{\theta^\circ}{360} \cdot 2\pi r, \quad A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{\theta^\circ}{360} \cdot \pi r^2 = \frac{1}{2}rs, \quad C = 2\pi r, \quad A = \pi r^2$$

$$\text{rad} = \text{deg} \cdot \frac{\pi}{180}, \quad \text{deg} = \text{rad} \cdot \frac{180}{\pi}, \quad \text{inscribed} = \frac{1}{2}\text{central}, \quad d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

4 · MINI THEORY SHEET

Equation of a circle. Group x -terms and y -terms, add $\left(\frac{\text{coefficient}}{2}\right)^2$ to *both sides* for each variable, then read h , k and r^2 . Signs flip: $(x+3)^2$ means $h = -3$. If a problem gives the radius, set the right-hand side equal to r^2 and solve for the constant.

Which formula for arcs and sectors?

Angle given in	Use
degrees	fraction $\frac{\theta}{360}$ of $2\pi r$ (arc) or of πr^2 (sector)
radians	$s = r\theta$ and $A = \frac{1}{2}r^2\theta$
arc and sector area known	$A = \frac{1}{2}rs$ gives r without knowing θ

Angles in a circle. Central angle = its arc (in degrees). Inscribed angle = $\frac{1}{2}$ its arc. Same arc $\Rightarrow$ equal inscribed angles. Draw radii to make isosceles triangles ($OA = OB = r$).

Tangents. Radius $\perp$ tangent gives a right triangle with legs r and the tangent length, and hypotenuse from the center to the outside point. Tangent lengths from one point are equal.

Radians. Convert with $\pi \leftrightarrow 180^\circ$. Co-terminal angles differ by 2π . A point at angle θ on a circle of radius r is $(r \cos \theta, r \sin \theta)$; moving clockwise subtracts the angle.

Speed shortcuts.

Situation	Shortcut
radius from $x^2 + y^2 + ax + by + c = 0$	$r^2 = \frac{a^2+b^2}{4} - c$
center from expanded form	$\left(-\frac{a}{2}, -\frac{b}{2}\right)$
circle tangent to an axis	the radius equals that center coordinate
diameter endpoints	center = midpoint, $r = \frac{1}{2} \cdot \text{distance}$
two sectors with equal area	$r_1^2\theta_1 = r_2^2\theta_2$
arc and radius given	$\theta = \frac{s}{r}$ rad, no degrees needed
tangent from an outside point	$\text{length}^2 = d^2 - r^2$
radius doubled/tripled	r^2 is multiplied by 4 or 9

5 · GUIDED EXAMPLES

Example 1 — Complete the square

The graph of $x^2 + y^2 + 6x - 8y - 11 = 0$ is a circle. Find its center and radius.

Group: $(x^2+6x)+(y^2-8y) = 11$. Add $\left(\frac{6}{2}\right)^2 = 9$ and $\left(\frac{8}{2}\right)^2 = 16$ to both sides: $(x+3)^2+(y-4)^2 = 11+9+16 = 36$. Center $(-3, 4)$, radius $\sqrt{36} = 6$.

Example 2 — Arc length and sector area

A circle has radius 9. A central angle measures 80° . Find the arc length and the sector area.

In radians, $80^\circ = 80 \cdot \frac{\pi}{180} = \frac{4\pi}{9}$. Arc: $s = r\theta = 9 \cdot \frac{4\pi}{9} = 4\pi$.

Sector: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(81)\frac{4\pi}{9} = 18\pi$. (Check: $\frac{80}{360} \cdot \pi(81) = 18\pi$.)

Example 3 — Tangent, radius and inscribed angle

$\overline{PA}$ is tangent to a circle with center O and radius 7 at A , and $OP = 25$. Also, an angle inscribed in the circle measures 35° . Find PA and the central angle on the same arc.

$OA \perp PA$, so $PA = \sqrt{25^2 - 7^2} = \sqrt{576} = 24$ (the triple 7-24-25).

Central angle $= 2 \cdot 35^\circ = 70^\circ$.

6 · PRACTICE SET — 48 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

The equation of a circle in the xy -plane is $(x - 3)^2 + (y + 5)^2 = 49$. Which of the following gives the center and the radius of the circle?

(A) center $(-3, 5)$, radius 49

(B) center $(3, -5)$, radius 7 ✓

(C) center $(-3, 5)$, radius 7

(D) center $(3, -5)$, radius 49

1. Compare with $(x - h)^2 + (y - k)^2 = r^2$: $h = 3$ and $k = -5$ (the sign flips), so the center is $(3, -5)$.

2. $r^2 = 49 \Rightarrow r = 7$.

Answer: (B)

2. EASY · NO CALCULATOR NEEDED

Which of the following is equal to $\frac{5\pi}{6}$ radians?

(A) 135°

(B) 300°

(C) 150° ✓

(D) 120°

1. Multiply by $\frac{180^\circ}{\pi}$: $\frac{5\pi}{6} \cdot \frac{180}{\pi} = 5 \cdot 30 = 150$.

2. So $\frac{5\pi}{6}$ radians $= 150^\circ$.

Answer: (C)

3. EASY · NO CALCULATOR NEEDED

A circle has a circumference of 20π . The area of the circle is $k\pi$. What is the value of k ?

1. $C = 2\pi r = 20\pi \Rightarrow r = 10$.

2. Area $= \pi r^2 = 100\pi$, so $k = 100$.

Answer: 100

4. EASY · NO CALCULATOR NEEDED

What is the value of $\sin \frac{5\pi}{6}$?

(A) $\frac{1}{2}$ ✓

(B) $-\frac{1}{2}$

(C) $\frac{\sqrt{3}}{2}$

(D) $-\frac{\sqrt{3}}{2}$

1. $\frac{5\pi}{6} = 150^\circ$, in Quadrant II where sine is positive.

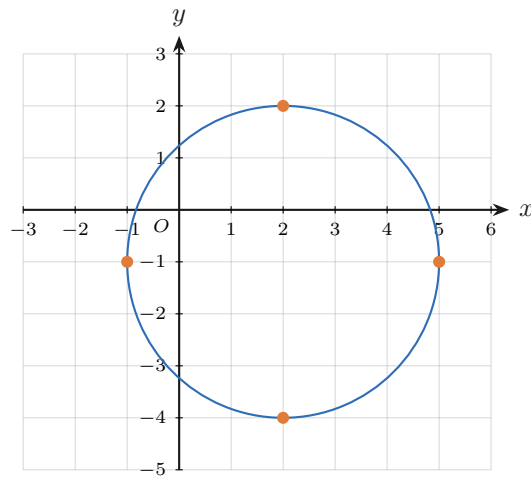
2. The reference angle is $180^\circ - 150^\circ = 30^\circ$, and $\sin 30^\circ = \frac{1}{2}$.

Answer: (A)

5.

EASY · DESMOS

The graph of a circle in the xy -plane is shown. Which of the following is an equation of the circle?



(A) $(x + 2)^2 + (y - 1)^2 = 9$

(B) $(x - 2)^2 + (y + 1)^2 = 3$

(C) $(x - 2)^2 + (y - 1)^2 = 9$

(D) $(x - 2)^2 + (y + 1)^2 = 9$ ✓

1. Read the horizontal diameter: the leftmost and rightmost grid points are $(-1, -1)$ and $(5, -1)$, so the center is $(2, -1)$ and the radius is 3.

2. $(x - 2)^2 + (y + 1)^2 = 3^2 = 9$.

Answer: (D)

6.

MEDIUM · NO CALCULATOR NEEDED

In the xy -plane, the graph of $x^2 + y^2 - 10x + 4y + 13 = 0$ is a circle. What is the radius of the circle?

1. Complete the square: $(x^2 - 10x + 25) + (y^2 + 4y + 4) = -13 + 25 + 4$.

2. $(x - 5)^2 + (y + 2)^2 = 16$.

3. $r = \sqrt{16} = 4$.

Answer: 4

7.

MEDIUM · NO CALCULATOR NEEDED

A circle has a radius of 12 centimeters. An arc of the circle has a central angle of 75° . What is the length, in centimeters, of the arc?

(A) 30π (B) 75π (C) 5π ✓(D) 10π

1. $\text{Arc} = \frac{75}{360} \cdot 2\pi(12)$.

2. $\frac{75}{360} = \frac{5}{24}$, so the arc is $\frac{5}{24} \cdot 24\pi = 5\pi$.

Answer: (C)

8.

MEDIUM · NO CALCULATOR NEEDED

Which of the following points lies on the circle with equation $(x + 1)^2 + (y - 4)^2 = 25$?

(A) $(1, -4)$ (B) $(2, 8)$ ✓(C) $(-2, 7)$ (D) $(3, 0)$

1. Substitute each point; the sum of the squares must be 25.

2. $(2, 8)$: $(2 + 1)^2 + (8 - 4)^2 = 9 + 16 = 25$. ✓

3. $(-2, 7)$ gives $1 + 9 = 10$; $(3, 0)$ gives $16 + 16 = 32$; $(1, -4)$ gives $4 + 64 = 68$.

Answer: (B)

9.

MEDIUM · NO CALCULATOR NEEDED

A circular sector has a radius of 6 and a central angle of $\frac{2\pi}{3}$ radians. The area of the sector is $k\pi$. What is the value of k ?

1. $A = \frac{1}{2}r^2\theta = \frac{1}{2}(36)\left(\frac{2\pi}{3}\right)$.

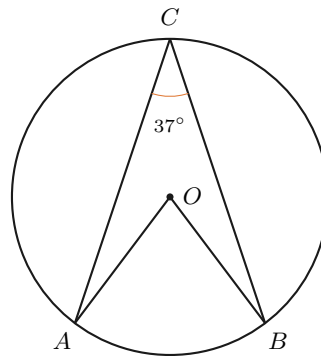
2. $A = 18 \cdot \frac{2\pi}{3} = 12\pi$, so $k = 12$.

Answer: 12

10.

MEDIUM · NO CALCULATOR NEEDED

In the figure, A , B and C lie on the circle with center O . What is the measure, in degrees, of $\angle AOB$?



(A) 37

(B) 143

(C) 106

(D) 74 ✓

1. $\angle ACB$ is an inscribed angle and $\angle AOB$ is the central angle on the same arc AB .

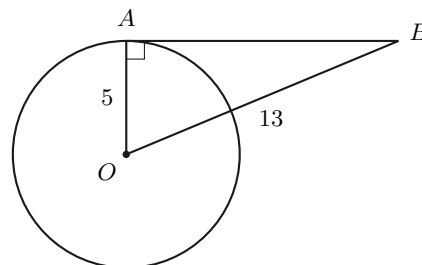
2. A central angle is twice the inscribed angle: $\angle AOB = 2(37^\circ) = 74^\circ$.

Answer: (D)

11.

MEDIUM · NO CALCULATOR NEEDED

In the figure, $\overline{AB}$ is tangent to the circle with center O at point A . What is the length of $\overline{AB}$?



(A) 12 ✓

(B) 8

(C) $\sqrt{194}$

(D) 18

1. A tangent is perpendicular to the radius at the point of tangency, so $\angle OAB = 90^\circ$.

2. $AB = \sqrt{OB^2 - OA^2} = \sqrt{13^2 - 5^2} = \sqrt{144} = 12$.

Answer: (A)

12.

MEDIUM · NO CALCULATOR NEEDED

A circle has a radius of 24. An arc of the circle has length 10π . What is the measure, in degrees, of the central angle that intercepts this arc?

1. Radian measure: $\theta = \frac{s}{r} = \frac{10\pi}{24} = \frac{5\pi}{12}$.

2. Degrees: $\frac{5\pi}{12} \cdot \frac{180}{\pi} = 75$.

Answer: 75

13.

MEDIUM · NO CALCULATOR NEEDED

The endpoints of a diameter of a circle in the xy -plane are $(-2, 3)$ and $(6, -5)$. Which of the following is an equation of the circle?

(A) $(x - 2)^2 + (y + 1)^2 = 16$

(B) $(x - 2)^2 + (y + 1)^2 = 32$ ✓

(C) $(x - 2)^2 + (y + 1)^2 = 128$

(D) $(x + 2)^2 + (y - 1)^2 = 64$

1. Center = midpoint = $\left(\frac{-2+6}{2}, \frac{3+(-5)}{2}\right) = (2, -1)$.

2. Diameter length squared: $(6 - (-2))^2 + (-5 - 3)^2 = 64 + 64 = 128$, so $r^2 = \frac{128}{4} = 32$.

3. $(x - 2)^2 + (y + 1)^2 = 32$.

Answer: (B)

14.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the graph of $x^2 + y^2 - 8x - 6y + k = 0$ is a circle that is tangent to the x -axis, where k is a constant. What is the value of k ?

1. Complete the square: $(x - 4)^2 + (y - 3)^2 = 25 - k$. The center is $(4, 3)$.

2. Tangent to the x -axis means the radius equals the center's distance to the axis: $r = 3$, so $r^2 = 9$.

3. $25 - k = 9 \Rightarrow k = 16$.

Answer: 16

15.

HARD · NO CALCULATOR NEEDED

In the xy -plane, two different circles are each tangent to both coordinate axes, lie in Quadrant IV, and pass through the point $(9, -2)$. What is the sum of the radii of the two circles?

1. A circle in Quadrant IV tangent to both axes has center $(r, -r)$ and radius r : $(x - r)^2 + (y + r)^2 = r^2$.
2. Substitute $(9, -2)$: $(9 - r)^2 + (-2 + r)^2 = r^2 \Rightarrow r^2 - 22r + 85 = 0$.
3. The two roots are the two radii, $r = 5$ and $r = 17$ (check: $(9 - 5)^2 + (-2 + 5)^2 = 25$ and $(9 - 17)^2 + (-2 + 17)^2 = 289$). Their sum is $5 + 17 = 22$ (the sum of the roots of $r^2 - 22r + 85 = 0$).

Answer: 22

16.

HARD · NO CALCULATOR NEEDED

The circle $(x + 2)^2 + (y - 3)^2 = 20$ is translated 4 units to the left and 5 units down. Then the radius of the translated circle is tripled, with the center unchanged. Which of the following is an equation of the resulting circle?

- (A) $(x - 2)^2 + (y - 8)^2 = 180$ (B) $(x + 6)^2 + (y + 2)^2 = 540$
(C) $(x + 6)^2 + (y + 2)^2 = 180$ ✓ (D) $(x + 6)^2 + (y + 2)^2 = 60$

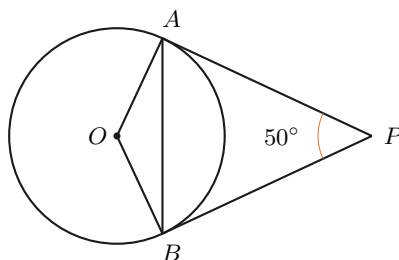
1. Translating left 4 and down 5 moves the center $(-2, 3)$ to $(-6, -2)$.
2. Tripling the radius multiplies r^2 by $3^2 = 9$: $r^2 = 20 \cdot 9 = 180$.
3. $(x + 6)^2 + (y + 2)^2 = 180$.

Answer: (C)

17.

HARD · NO CALCULATOR NEEDED

In the figure, $\overline{PA}$ and $\overline{PB}$ are tangent to the circle with center O at A and B , and $\angle APB = 50^\circ$. What is the measure, in degrees, of $\angle OAB$?



(A) 25 ✓

(B) 40

(C) 50

(D) 65

1. $\angle OAP = \angle OBP = 90^\circ$ (tangent $\perp$ radius). In quadrilateral $OAPB$: $\angle AOB = 360 - 90 - 90 - 50 = 130^\circ$.

2. $OA = OB$ (radii), so $\triangle OAB$ is isosceles with base angles $\angle OAB = \angle OBA$.

3. $\angle OAB = \frac{180 - 130}{2} = 25$.

Answer: (A)

18.

HARD · NO CALCULATOR NEEDED

A circular sector has an area of 40 square centimeters and an arc length of 10 centimeters. What is the radius, in centimeters, of the circle?

1. For a sector, $A = \frac{1}{2}rs$ (a triangle-like formula using the arc length s).

2. $40 = \frac{1}{2}r(10) = 5r \Rightarrow r = 8$.

3. Check: $\theta = \frac{s}{r} = \frac{10}{8} = 1.25$ rad, and $\frac{1}{2}r^2\theta = \frac{1}{2}(64)(1.25) = 40$.

Answer: 8

19.

HARD · NO CALCULATOR NEEDED

For an angle θ with $\pi < \theta < \frac{3\pi}{2}$, $\sin \theta = -\frac{5}{13}$. What is the value of $\tan \theta$?

(A) $-\frac{5}{12}$

(B) $\frac{12}{5}$

(C) $-\frac{12}{5}$

(D) $\frac{5}{12}$ ✓

1. $\pi < \theta < \frac{3\pi}{2}$ is Quadrant III: both sine and cosine are negative.

2. $\cos \theta = -\sqrt{1 - \left(\frac{5}{13}\right)^2} = -\frac{12}{13}$.

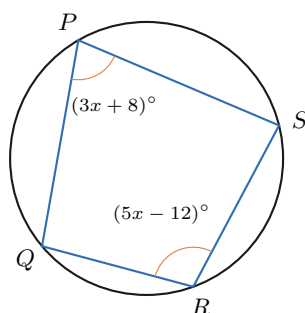
3. $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{-5/13}{-12/13} = \frac{5}{12}$ (positive in Quadrant III).

Answer: (D)

20.

HARD · NO CALCULATOR NEEDED

Quadrilateral $PQRS$ is inscribed in a circle. The measures of angles P and R are $(3x + 8)^\circ$ and $(5x - 12)^\circ$, respectively. What is the measure, in degrees, of angle R ?



(A) 115

(B) 77

(C) 103 ✓

(D) 23

1. Angles P and R are opposite angles of an inscribed quadrilateral. Each intercepts one of the two arcs that together make 360° , so $\angle P + \angle R = \frac{360^\circ}{2} = 180^\circ$.

2. $(3x + 8) + (5x - 12) = 180 \Rightarrow 8x - 4 = 180 \Rightarrow x = 23$.

3. $\angle R = 5(23) - 12 = 103$. (Check: $\angle P = 3(23) + 8 = 77$ and $77 + 103 = 180$.)

Answer: (C)

21.

HARD · CALCULATOR

A circular running track has a radius of 0.12 kilometers. A runner starts at one point on the track and runs along the track through a central angle of 200° . To the nearest meter, how far does the runner run?

1. Convert the radius: $0.12 \text{ km} = 120 \text{ m}$.

2. Arc length $= \frac{200}{360} \cdot 2\pi(120) = \frac{5}{9} \cdot 240\pi = \frac{400\pi}{3} \approx 418.9 \text{ m}$.

3. To the nearest meter: 419.

Answer: 419

22.

HARD · NO CALCULATOR NEEDED

A circular pizza with a radius of 9 inches is cut into slices, and one slice has a central angle of 40° . A second circular pizza has a radius of 12 inches. A slice of the second pizza has the same area as the 40° slice of the first pizza. What is the central angle, in degrees, of the slice of the second pizza?

(A) 71.1

(B) 22.5 ✓

(C) 30

(D) 53.3

1. Sector area is $\frac{\theta}{360}\pi r^2$, so equal areas give $\theta_1 r_1^2 = \theta_2 r_2^2$.

2. $40 \cdot 9^2 = \theta_2 \cdot 12^2 \Rightarrow \theta_2 = \frac{40 \cdot 81}{144}$.

3. $\theta_2 = 22.5$.

Answer: (B)

23.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the graph of $x^2 + y^2 - 6x + 2ky - 39 = 0$ is a circle whose center lies on the line $y = x + 1$, where k is a constant. What is the radius of the circle?

1. Complete the square: $(x - 3)^2 + (y + k)^2 = 39 + 9 + k^2$. The center is $(3, -k)$.

2. The center is on $y = x + 1$: $-k = 3 + 1 \Rightarrow k = -4$.

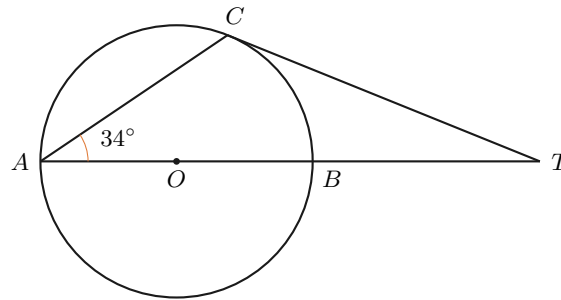
3. $r^2 = 39 + 9 + (-4)^2 = 64$, so $r = 8$.

Answer: 8

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, $\overline{AB}$ is a diameter of the circle with center O , point C lies on the circle, and line CT is tangent to the circle at C with T on line AB . What is the measure, in degrees, of $\angle CTA$?



(A) 34

(B) 56

(C) 68

(D) 22 ✓

- $\angle CAB = 34^\circ$ is an inscribed angle on arc CB , so the central angle is $\angle COB = 2(34^\circ) = 68^\circ$.
- The tangent is perpendicular to the radius: $\angle OCT = 90^\circ$.
- In right triangle OCT : $\angle CTO = 90^\circ - 68^\circ = 22^\circ$.
- $\angle CTA$ is the same angle as $\angle CTO$ (points A, O, T are collinear), so it measures 22° .

Answer: (D)

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

A circular sector has a perimeter of 20 centimeters (two radii plus the arc) and an area of 24 square centimeters. The central angle of the sector is θ radians. What is the sum of all possible values of θ ?

- Let the radius be r and the arc be $r\theta$. Perimeter: $2r + r\theta = 20$, so the arc is $20 - 2r$.
- Area = $\frac{1}{2}r(\text{arc}) = \frac{1}{2}r(20 - 2r) = 10r - r^2 = 24 \Rightarrow r^2 - 10r + 24 = 0 \Rightarrow r = 4$ or $r = 6$.
- $\theta = \frac{20 - 2r}{r}$: for $r = 4$, $\theta = \frac{12}{4} = 3$; for $r = 6$, $\theta = \frac{8}{6} = \frac{4}{3}$.
- Both are valid (each is less than 2π). Sum = $3 + \frac{4}{3} = \frac{13}{3}$.

Answer: $\frac{13}{3}$

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

The line $3x + 4y = k$, where k is a positive constant, is tangent to the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ in the xy -plane. What is the value of k ?

(A) 19 ✓

(B) 25

(C) 31

(D) -31

1. Complete the square: $(x - 2)^2 + (y + 3)^2 = 12 + 4 + 9 = 25$. Center $(2, -3)$, radius 5.
2. The radius to a point of tangency is perpendicular to the tangent line. The line $3x + 4y = k$ has direction normal $(3, 4)$, so the radius points along $\pm(3, 4)/5$ and the tangent points are $(2, -3) \pm 5 \cdot \frac{(3, 4)}{5} = (5, 1)$ and $(-1, -7)$.
3. $k = 3(5) + 4(1) = 19$ or $k = 3(-1) + 4(-7) = -31$.
4. k is positive, so $k = 19$.

Answer: (A)**27.**

SAT CHALLENGE · NO CALCULATOR NEEDED

Circle A is given by $x^2 + y^2 - 2x - 4y - 11 = 0$ and circle B is given by $x^2 + y^2 - 14x + 12y + c = 0$ in the xy -plane, where c is a constant. The two circles are externally tangent (they touch at exactly one point and neither lies inside the other). What is the value of c ?

1. Circle A : $(x - 1)^2 + (y - 2)^2 = 16$, center $(1, 2)$, radius 4.
2. Circle B : $(x - 7)^2 + (y + 6)^2 = 85 - c$, center $(7, -6)$, radius $r_B = \sqrt{85 - c}$.
3. Distance between centers: $\sqrt{6^2 + 8^2} = 10$. External tangency: $10 = 4 + r_B \Rightarrow r_B = 6$.
4. $85 - c = 36 \Rightarrow c = 49$.

Answer: 49

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

A point (a, b) lies on the unit circle at angle θ radians in standard position, where $0 < \theta < 2\pi$ and $ab > 0$. Which of the following must be true?

I. $\tan \theta > 0$ II. $\sin(\theta + \pi) \cos(\theta + \pi) > 0$ III. $\sin \theta > 0$

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II, and III

1. $a = \cos \theta$ and $b = \sin \theta$, and $ab > 0$ means sine and cosine have the same sign: Quadrant I or Quadrant III.

2. I: $\tan \theta = \frac{b}{a} > 0$ in both cases. True.

3. II: adding π negates both: $\sin(\theta + \pi) \cos(\theta + \pi) = (-b)(-a) = ab > 0$. True.

4. III: in Quadrant III, $\sin \theta < 0$, so III need not be true.

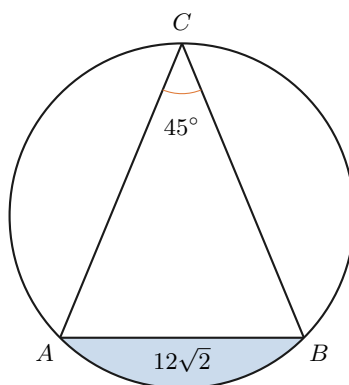
5. I and II only.

Answer: (B)

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, A , B and C lie on a circle, $\angle ACB = 45^\circ$, and $AB = 12\sqrt{2}$. The area of the shaded region, bounded by chord $\overline{AB}$ and the arc AB that does not contain C , is $a\pi - b$, where a and b are positive integers. What is the value of $a + b$?



1. The inscribed angle 45° gives the central angle $\angle AOB = 90^\circ$ (with center O).
2. $\triangle AOB$ is an isosceles right triangle with hypotenuse $AB = 12\sqrt{2}$, so $r\sqrt{2} = 12\sqrt{2}$ and $r = 12$.
3. Sector $AOB = \frac{90}{360}\pi(12)^2 = 36\pi$; triangle $AOB = \frac{1}{2}(12)(12) = 72$.
4. Shaded region $= 36\pi - 72$, so $a = 36$, $b = 72$ and $a + b = 108$.

Answer: 108

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

A point starts at $(6, 0)$ and moves clockwise along the circle $x^2 + y^2 = 36$ a distance of 11π . What are the coordinates of the point where it stops?

(A) $(3\sqrt{3}, -3)$

(B) $(-3\sqrt{3}, 3)$

(C) $(3\sqrt{3}, 3)$ ✓

(D) $(3, 3\sqrt{3})$

1. The central angle is $\theta = \frac{s}{r} = \frac{11\pi}{6}$ radians, measured clockwise, i.e. $-\frac{11\pi}{6}$ in standard position.

2. $-\frac{11\pi}{6} + 2\pi = \frac{\pi}{6}$, so the final position is at angle 30° .

3. $(x, y) = \left(6 \cos \frac{\pi}{6}, 6 \sin \frac{\pi}{6}\right) = \left(6 \cdot \frac{\sqrt{3}}{2}, 6 \cdot \frac{1}{2}\right) = (3\sqrt{3}, 3)$.

Answer: (C)**31.**

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, a circle passes through the points $(-1, 2)$, $(7, 2)$ and $(6, 3)$. Which of the following is an equation of the circle?

(A) $(x - 3)^2 + (y + 1)^2 = 25$ ✓

(B) $(x - 3)^2 + (y - 2)^2 = 16$

(C) $(x - 3)^2 + (y + 1)^2 = 5$

(D) $(x - 3)^2 + (y - 1)^2 = 17$

1. The points $(-1, 2)$ and $(7, 2)$ form a horizontal chord, so the center lies on the vertical line $x = \frac{-1+7}{2} = 3$. (It is *not* the midpoint $(3, 2)$; the chord need not be a diameter.)

2. Let the center be $(3, k)$. Equal distances to $(7, 2)$ and $(6, 3)$: $16 + (2 - k)^2 = 9 + (3 - k)^2 \Rightarrow 20 - 4k = 18 - 6k \Rightarrow k = -1$.

3. $r^2 = (7 - 3)^2 + (2 + 1)^2 = 16 + 9 = 25$.

4. Equation: $(x - 3)^2 + (y + 1)^2 = 25$. Check $(6, 3)$: $9 + 16 = 25$.

Answer: (A)

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^2 + y^2 - 6x + 8y = k$$

In the xy -plane, the graph of the equation above, where k is a constant, is a circle that is tangent to the line $4x - 3y = -11$. What is the value of k ?

1. Complete the square: $(x - 3)^2 + (y + 4)^2 = k + 25$. The center is $(3, -4)$ and $r^2 = k + 25$.

2. Distance from the center to $4x - 3y + 11 = 0$: $\frac{|4(3) - 3(-4) + 11|}{\sqrt{16 + 9}} = \frac{35}{5} = 7$.

3. Tangency means $r = 7$, so $r^2 = 49 = k + 25$.

4. $k = 24$.

Answer: 24

33.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$(x - a - 2)^2 + (y - 2a + 1)^2 = 25$$

In the xy -plane, the circle above is defined for each real number a . Which of the following points lies on the circle for every real value of a ?

(A) $(a + 5, 2a + 4)$

(B) $(2a + 5, 2a + 3)$

(C) $(a - 1, 2a + 2)$

(D) $(a + 5, 2a + 3)$ ✓

1. The circle has center $(a + 2, 2a - 1)$ and radius 5. For a point (x, y) , compute the offsets $x - (a + 2)$ and $y - (2a - 1)$ and see whether the a -terms cancel.

2. $(a + 5, 2a + 3)$: offsets 3 and 4, and $3^2 + 4^2 = 25$. ✓

3. $(a + 5, 2a + 4)$: offsets 3, 5, sum of squares 34. $(2a + 5, 2a + 3)$: first offset $a + 3$ still depends on a . $(a - 1, 2a + 2)$: offsets $-3, 3$, sum of squares 18.

4. Only $(a + 5, 2a + 3)$ works.

Answer: (D)

34.

SAT CHALLENGE · CALCULATOR

In triangle ABC , angle C is a right angle and angle A measures 30° . The circle inscribed in the triangle has radius 5. What is the length of $\overline{AC}$?

(A) $10\sqrt{3} - 10$

(B) $5 + 5\sqrt{3}$

(C) $15 + 5\sqrt{3}$ ✓

(D) $10 + 10\sqrt{3}$

1. Let $BC = s$ (opposite 30°). Then $AC = s\sqrt{3}$ and $AB = 2s$.

2. Inradius of a right triangle: $r = \frac{BC + AC - AB}{2} = \frac{s + s\sqrt{3} - 2s}{2} = \frac{s(\sqrt{3} - 1)}{2} = 5$.

3. $s = \frac{10}{\sqrt{3} - 1} = \frac{10(\sqrt{3} + 1)}{2} = 5\sqrt{3} + 5$.

4. $AC = s\sqrt{3} = 5\sqrt{3}(\sqrt{3} + 1) = 15 + 5\sqrt{3}$.

5. (Area check: $\frac{1}{2} \cdot s \cdot s\sqrt{3}$ divided by the semiperimeter equals 5.)

Answer: (C)

35.

SAT CHALLENGE · CALCULATOR

$$(x - 1)^2 + (y + 3)^2 = 40$$

$$x^2 + y^2 - 4kx + 2ky = 9$$

In the xy -plane, the two equations above define circles. The center of the second circle lies on the first circle. If k is a positive constant, what is the value of k ?

(A) $1 + 2\sqrt{7}$

(B) $1 + \sqrt{7}$ ✓

(C) $-1 + \sqrt{7}$

(D) $2 + \sqrt{7}$

1. The second circle has center $(2k, -k)$, since $x^2 - 4kx = (x - 2k)^2 - 4k^2$ and $y^2 + 2ky = (y + k)^2 - k^2$.

2. Substitute into the first equation: $(2k - 1)^2 + (-k + 3)^2 = 40$.

3. $4k^2 - 4k + 1 + k^2 - 6k + 9 = 40 \Rightarrow 5k^2 - 10k - 30 = 0 \Rightarrow k^2 - 2k - 6 = 0$.

4. $k = \frac{2 \pm \sqrt{4 + 24}}{2} = \frac{2 \pm 2\sqrt{7}}{2} = 1 \pm \sqrt{7}$.

5. Since $k > 0$, $k = 1 + \sqrt{7}$ (because $1 - \sqrt{7} < 0$).

Answer: (B)

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, a circle has equation $x^2 + y^2 + Dx + Ey + F = 0$, where D , E , and F are constants. The circle passes through the points $(2, 7)$ and $(10, 7)$, has radius 5, and has its center below the line $y = 7$. What is the value of $E + F$?

- The points $(2, 7)$ and $(10, 7)$ have the same y -coordinate, so the center lies on the perpendicular bisector $x = 6$. The center is $\left(-\frac{D}{2}, -\frac{E}{2}\right)$, so $D = -12$.
- Center $(6, k)$ with distance 5 to $(2, 7)$: $16 + (7 - k)^2 = 25 \Rightarrow (7 - k)^2 = 9 \Rightarrow k = 4$ or $k = 10$. Below $y = 7$ means $k = 4$.
- $E = -2k = -8$. Also $F = h^2 + k^2 - r^2 = 36 + 16 - 25 = 27$.
- $E + F = -8 + 27 = 19$.
- (Check with $(2, 7)$: $4 + 49 - 24 - 56 + 27 = 0$.)

Answer: 19**37.**

SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^2 + y^2 - 4x + 6y - 12 = 0$$

In the xy -plane, the graph of the equation above is a circle. Line ℓ has slope $\frac{3}{4}$ and is tangent to the circle at a point with a positive y -coordinate. What is the y -intercept of ℓ ?

(A) $-\frac{1}{3}$

(B) $-\frac{43}{4}$

(C) 1

(D) $\frac{7}{4}$ ✓

- Complete the squares: $(x - 2)^2 + (y + 3)^2 = 25$. Center $(2, -3)$, radius 5.
- The radius to the point of tangency is perpendicular to ℓ , so it has slope $-\frac{4}{3}$. A radius of length 5 with slope $-\frac{4}{3}$ is $\pm(3, -4)$ from the center.
- Tangency points: $(2 + 3, -3 - 4) = (5, -7)$ and $(2 - 3, -3 + 4) = (-1, 1)$. Only $(-1, 1)$ has a positive y -coordinate.
- Line through $(-1, 1)$ with slope $\frac{3}{4}$: $y - 1 = \frac{3}{4}(x + 1)$, so $y = \frac{3}{4}x + \frac{7}{4}$.
- y -intercept $= \frac{7}{4}$. (Other options: $-\frac{43}{4}$ is the other tangent line, 1 is the y -coordinate of the tangency point, $-\frac{1}{3}$ uses slope $-\frac{4}{3}$.)

Answer: (D)

38.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the circle with equation $(x - 7)^2 + (y + 2)^2 = 9$ has two distinct tangent lines that pass through the point $(2, -2)$. What is the product of the slopes of these two lines?

1. The point $(2, -2)$ is level with the centre $(7, -2)$, so the situation is symmetric about the horizontal line $y = -2$: the two tangent slopes are m and $-m$, and the product is $-m^2$.
2. Distance from the point to the centre: $d = 5$; radius $r = 3$. The tangent makes angle θ with the line to the centre where $\sin \theta = \frac{3}{5}$, so $\tan \theta = \frac{3}{4}$ and $m^2 = \frac{9}{16}$.
3. Check by the discriminant: substituting $y = -2 + m(x - 2)$ gives $(1 + m^2)x^2 - (14 + 4m^2)x + (40 + 4m^2) = 0$; its discriminant is $36 - 64m^2$, which is 0 for $m^2 = \frac{9}{16}$.
4. Product $= -\frac{9}{16}$.

Answer: $-9/16$ **39.**

SAT CHALLENGE · CALCULATOR

In the xy -plane, a circle lies in the first quadrant and is tangent to both the x -axis and the y -axis. The line $3x + 4y = 20$ intersects the circle at two points 8 units apart. What is the sum of all possible values of the radius of the circle?

1. Tangent to both axes in the first quadrant: centre (r, r) and radius r .
2. Distance from the centre to the line: $d = \frac{|3r + 4r - 20|}{5} = \frac{|7r - 20|}{5}$.
3. The perpendicular from the centre bisects the chord, so $d^2 + 4^2 = r^2$: $\frac{(7r - 20)^2}{25} + 16 = r^2$.
4. $49r^2 - 280r + 400 + 400 = 25r^2 \Rightarrow 24r^2 - 280r + 800 = 0 \Rightarrow 3r^2 - 35r + 100 = 0$.
5. Roots $r = 5$ and $r = \frac{20}{3}$, both positive and both valid (check $d = 3$ and $d = \frac{16}{3}$). Their sum is $\frac{35}{3}$ (Vieta: $-\frac{-35}{3}$).

Answer: $35/3$

40.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the center of a circle lies on the y -axis. The circle passes through the points $(6, 1)$ and $(-2, 5)$. Which of the following is an equation of the circle?

(A) $x^2 + (y + 1)^2 = 40$ ✓

(B) $x^2 + (y - 1)^2 = 40$

(C) $x^2 + (y + 1)^2 = 20$

(D) $x^2 + (y + 1)^2 = 44$

1. The centre is $(0, k)$. Equal distances to the two points: $6^2 + (1 - k)^2 = (-2)^2 + (5 - k)^2$.
2. Expand: $36 + 1 - 2k + k^2 = 4 + 25 - 10k + k^2$, so $37 - 2k = 29 - 10k$.
3. $8k = -8 \Rightarrow k = -1$. Centre $(0, -1)$.
4. $r^2 = 6^2 + (1 - (-1))^2 = 36 + 4 = 40$. Check with the other point: $(-2)^2 + (5 + 1)^2 = 4 + 36 = 40$.
5. Equation: $x^2 + (y + 1)^2 = 40$.

Answer: (A)

41.

SAT CHALLENGE · NO CALCULATOR NEEDED

Triangle ABC is inscribed in a circle with circumference 52π . Side AC has length 52 and side AB has length 48. What is the ratio of the length of the shortest side of the triangle to the radius of the circle?

(A) $\frac{5}{13}$

(B) $\frac{10}{13}$ ✓

(C) $\frac{24}{13}$

(D) $\frac{5}{12}$

1. Circumference $= 2\pi r = 52\pi$, so the radius is $r = 26$ and the diameter is 52.
2. $AC = 52$ equals the diameter, so $\overline{AC}$ is a diameter and the angle at B is a right angle (inscribed in a semicircle).
3. Pythagorean theorem: $BC = \sqrt{52^2 - 48^2} = \sqrt{(52 - 48)(52 + 48)} = \sqrt{400} = 20$.
4. The sides are 20, 48, 52; the shortest is $BC = 20$.
5. Ratio to the radius: $\frac{20}{26} = \frac{10}{13}$. (Dividing by the diameter would give $\frac{5}{13}$.)

Answer: (B)

42.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the circle $x^2 + y^2 + 8x - 2y - 8 = 0$ and the line $5x + 12y = 44$ intersect at two points, P and Q . What is the length of $\overline{PQ}$?

(A) 8

(B) 3

(C) 6 ✓

(D) $2\sqrt{21}$

1. Complete the squares: $(x + 4)^2 + (y - 1)^2 = 8 + 16 + 1 = 25$. Center $(-4, 1)$, radius $r = 5$.

2. Distance from the center to the line $5x + 12y - 44 = 0$: $d = \frac{|5(-4) + 12(1) - 44|}{\sqrt{25 + 144}} = \frac{52}{13} = 4$.

3. The perpendicular from the center bisects the chord, forming a right triangle with legs $d = 4$ and $\frac{PQ}{2}$ and hypotenuse $r = 5$: $\frac{PQ}{2} = \sqrt{25 - 16} = 3$.

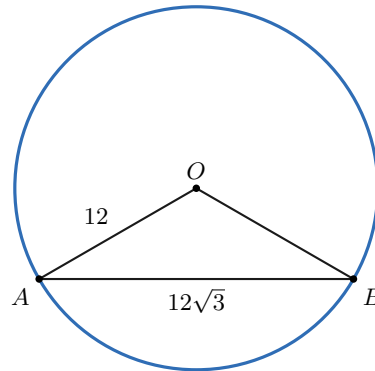
4. $PQ = 2 \cdot 3 = 6$. (3 is the half-chord; $2\sqrt{21} = 2\sqrt{25 - 4}$ comes from using $d = 2$; $8 = 2d$.)

Answer: (C)

43.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the circle shown, O is the center, the radius is 12, and chord $\overline{AB}$ has length $12\sqrt{3}$. What is the area of the region bounded by chord $\overline{AB}$ and the minor arc AB ?

(A) $48\pi - 36\sqrt{3}$ ✓(B) 48π (C) $96\pi + 36\sqrt{3}$ (D) $24\pi - 36\sqrt{3}$

- Triangle OAB has $OA = OB = 12$. Drop the perpendicular from O to $\overline{AB}$: it bisects the chord into two segments of length $6\sqrt{3}$.
- In each half-triangle, $\sin\left(\frac{\angle AOB}{2}\right) = \frac{6\sqrt{3}}{12} = \frac{\sqrt{3}}{2}$, so $\frac{\angle AOB}{2} = 60^\circ$ and $\angle AOB = 120^\circ$.
- Sector area: $\frac{120}{360}\pi(12)^2 = 48\pi$.
- Triangle area: $\frac{1}{2}(12)(12)\sin 120^\circ = 72 \cdot \frac{\sqrt{3}}{2} = 36\sqrt{3}$.
- Segment = sector – triangle = $48\pi - 36\sqrt{3}$. (Option 48π omits the triangle; $96\pi + 36\sqrt{3}$ is the major segment.)

Answer: (A)

44.

SAT CHALLENGE · NO CALCULATOR NEEDED

A circular clock face has radius 15 centimeters. A shaded sector of the clock face has area 45π square centimeters. What is the perimeter, in centimeters, of the shaded sector (the arc together with the two bounding radii)?

(A) $15 + 6\pi$

(B) $30 + 3\pi$

(C) $30 + 45\pi$

(D) $30 + 6\pi$ ✓

1. Area of the whole face: $\pi(15)^2 = 225\pi$. The sector is $\frac{45\pi}{225\pi} = \frac{1}{5}$ of the circle.

2. Circumference: $2\pi(15) = 30\pi$, so the arc length is $\frac{1}{5} \cdot 30\pi = 6\pi$.

3. Perimeter of the sector = arc + two radii = $6\pi + 2(15) = 30 + 6\pi$.

4. (Check with $A = \frac{1}{2}r^2\theta$: $45\pi = \frac{1}{2}(225)\theta \Rightarrow \theta = \frac{2\pi}{5}$, and arc = $r\theta = 6\pi$.)

Answer: (D)

45.

SAT CHALLENGE · NO CALCULATOR NEEDED

Segment $\overline{AB}$ is a diameter of a circle and has length 20. Point C lies on the circle, and point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 8$. If $AD < DB$, what is the value of $\frac{AC}{CB}$?

(A) $\frac{1}{4}$

(B) $\frac{2}{5}$

(C) $\frac{1}{2}$ ✓

(D) $\frac{4}{5}$

1. C lies on a circle with diameter AB , so $\angle ACB = 90^\circ$ and $\overline{CD}$ is the altitude to the hypotenuse.

2. $CD^2 = AD \cdot DB$ and $AD + DB = 20$: $AD \cdot DB = 64$. Solving $x(20 - x) = 64$ gives $x = 4$ or 16; with $AD < DB$, $AD = 4$ and $DB = 16$.

3. $AC^2 = AD \cdot AB = 4 \cdot 20 = 80$ and $CB^2 = DB \cdot AB = 16 \cdot 20 = 320$.

4. $\frac{AC}{CB} = \sqrt{\frac{80}{320}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$. (Note $\frac{AD}{DB} = \frac{1}{4}$ is the ratio of the segments, not of the legs.)

Answer: (C)

46.

SAT CHALLENGE · CALCULATOR

A circle has area 48π . A square and an equilateral triangle are each inscribed in the circle (all vertices on the circle). By how much does the area of the square exceed the area of the triangle?

(A) $96 - 27\sqrt{3}$

(B) $96 - 36\sqrt{3}$ ✓

(C) $96 - 12\sqrt{3}$

(D) $192 - 36\sqrt{3}$

1. $\pi r^2 = 48\pi$, so $r^2 = 48$ and $r = 4\sqrt{3}$.

2. Square: its diagonal is the diameter, $2r = 8\sqrt{3}$. Area = $\frac{d^2}{2} = \frac{192}{2} = 96$.

3. Triangle: the side is $r\sqrt{3} = 12$, so the area is $\frac{\sqrt{3}}{4} \cdot 12^2 = 36\sqrt{3}$.

4. Difference: $96 - 36\sqrt{3}$.

Answer: (B)

47.

SAT CHALLENGE · DESMOS

In the xy -plane, the circle $x^2 + y^2 + 8x - 2y = 8$ and the line $5x - 12y = k$, where k is a constant, intersect at exactly one point. What is the greatest possible value of k ?

(A) -97

(B) 65

(C) 32

(D) 33 ✓

1. Complete the square: $(x + 4)^2 + (y - 1)^2 = 8 + 16 + 1 = 25$. Centre $(-4, 1)$, radius 5.

2. One intersection point means the line is tangent, so the distance from the centre to the line equals 5: $\frac{|5(-4) - 12(1) - k|}{\sqrt{5^2 + 12^2}} = 5$.

3. $\frac{|-32 - k|}{13} = 5 \Rightarrow |k + 32| = 65$.

4. $k + 32 = 65$ gives $k = 33$; $k + 32 = -65$ gives $k = -97$. The greatest is 33.

5. (Option -97 is the other case; 65 and 32 come from not finishing the absolute-value equation.)

Answer: (D)

48.

SAT CHALLENGE · NO CALCULATOR NEEDED

In a circle with center O , chord $\overline{AB}$ has $\angle AOB = 120^\circ$ and lies at a perpendicular distance of 3 from O . The lines tangent to the circle at A and at B intersect at point P . What is the length of $\overline{PA}$?

(A) $6\sqrt{3}$ ✓

(B) 12

(C) $3\sqrt{3}$

(D) 6

1. Triangle OAB is isosceles with apex angle 120° . The perpendicular from O to AB bisects the apex angle, giving a right triangle with a 60° angle at O , so $3 = r \cos 60^\circ$ and $r = 6$.
2. A tangent is perpendicular to the radius, so $\angle OAP = 90^\circ$. By symmetry OP bisects $\angle AOB$, so $\angle AOP = 60^\circ$.
3. $PA = r \tan 60^\circ = 6\sqrt{3}$. (Equal tangent segments: $PB = PA$.)
4. (Option $3\sqrt{3}$ treats the distance 3 as the radius; 6 is r ; 12 is OP .)

Answer: (A)